

SAFETY DATA SHEET

MIXED ALCOHOL SOLVENT

According to Regulation (EC) No 1907/2006, Annex II, as amended. Commission Regulation (EU) No 2015/830 of 28 May 2015.

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Product name MIXED ALCOHOL SOLVENT

Product number 2284

1.2. Relevant identified uses of the substance or mixture and uses advised against

Identified uses Laboratory reagent.

Uses advised against No specific uses advised against are identified.

1.3. Details of the supplier of the safety data sheet

Supplier Reagent Chemical Services
 11b - 13 Aston Fields Road
 Whitehouse Industrial Estate
 Runcorn
 Cheshire WA7 3DL
 T: 01928 716903 (08.30 - 17.00)
 F: 01928 716425
 E: info@reagent.co.uk

1.4. Emergency telephone number

Emergency telephone OHES Environmental Ltd 24-7
 Tel. 0333 333 9939 (24 hour)

SECTION 2: Hazards identification

2.1. Classification of the substance or mixture

Classification (EC 1272/2008)

Physical hazards Flam. Liq. 2 - H225

Health hazards Acute Tox. 3 - H301 Acute Tox. 3 - H311 Skin Irrit. 2 - H315 Eye Dam. 1 - H318 STOT SE 1 - H370 STOT SE 3 - H335, H336

Environmental hazards Not Classified

2.2. Label elements

Hazard pictograms



Signal word

Danger

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Hazard statements	H225 Highly flammable liquid and vapour. H301+H311 Toxic if swallowed or in contact with skin. H315 Causes skin irritation. H318 Causes serious eye damage. H370 Causes damage to organs . H335 May cause respiratory irritation. H336 May cause drowsiness or dizziness.
Precautionary statements	P210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. P260 Do not breathe vapour/ spray. P271 Use only outdoors or in a well-ventilated area. P280 Wear protective gloves/ protective clothing/ eye protection/ face protection. P301+P310 IF SWALLOWED: Immediately call a POISON CENTER/ doctor. P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
Contains	BUTANOL-norm, METHANOL
Supplementary precautionary statements	P240 Ground and bond container and receiving equipment. P241 Use explosion-proof electrical equipment. P242 Use non-sparking tools. P243 Take action to prevent static discharges. P261 Avoid breathing vapour/ spray. P264 Wash contaminated skin thoroughly after handling. P270 Do not eat, drink or smoke when using this product. P302+P352 IF ON SKIN: Wash with plenty of water. P303+P361+P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower. P304+P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing. P308+P311 IF exposed or concerned: Call a POISON CENTER or doctor. P312 Call a POISON CENTRE/doctor if you feel unwell. P321 Specific treatment (see medical advice on this label). P330 Rinse mouth. P332+P313 If skin irritation occurs: Get medical advice/ attention. P361+P364 Take off immediately all contaminated clothing and wash it before reuse. P362+P364 Take off contaminated clothing and wash it before reuse. P370+P378 In case of fire: Use foam, carbon dioxide, dry powder or water fog to extinguish. P403+P233 Store in a well-ventilated place. Keep container tightly closed. P403+P235 Store in a well-ventilated place. Keep cool. P405 Store locked up. P501 Dispose of contents/ container in accordance with national regulations.

2.3. Other hazards

This product does not contain any substances classified as PBT or vPvB.

SECTION 3: Composition/information on ingredients

3.2. Mixtures

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BUTANOL-norm			60-100%
CAS number: 71-36-3	EC number: 200-751-6	REACH registration number: 01-2119484630-38-XXXX	
Classification Flam. Liq. 3 - H226 Acute Tox. 4 - H302 Skin Irrit. 2 - H315 Eye Dam. 1 - H318 STOT SE 3 - H335, H336			

METHANOL			30-60%
CAS number: 67-56-1	EC number: 200-659-6	REACH registration number: 01-2119433307-44-XXXX	
Classification Flam. Liq. 2 - H225 Acute Tox. 3 - H301 Acute Tox. 3 - H311 Acute Tox. 3 - H331 STOT SE 1 - H370			

The Full Text for all R-Phrases and Hazard Statements are Displayed in Section 16.

SECTION 4: First aid measures

4.1. Description of first aid measures

General information	Get medical attention immediately. Show this Safety Data Sheet to the medical personnel.
Inhalation	Remove affected person from source of contamination. Move affected person to fresh air and keep warm and at rest in a position comfortable for breathing. Maintain an open airway. Loosen tight clothing such as collar, tie or belt. When breathing is difficult, properly trained personnel may assist affected person by administering oxygen. Place unconscious person on their side in the recovery position and ensure breathing can take place.
Ingestion	Rinse mouth thoroughly with water. Remove any dentures. Stop if the affected person feels sick as vomiting may be dangerous. Do not induce vomiting unless under the direction of medical personnel. If vomiting occurs, the head should be kept low so that vomit does not enter the lungs. Never give anything by mouth to an unconscious person. Move affected person to fresh air and keep warm and at rest in a position comfortable for breathing. Place unconscious person on their side in the recovery position and ensure breathing can take place. Maintain an open airway. Loosen tight clothing such as collar, tie or belt.
Skin contact	It is important to remove the substance from the skin immediately. Take off immediately all contaminated clothing. Remove contamination with soap and water or recognised skin cleansing agent. Get medical attention.
Eye contact	Rinse immediately with plenty of water. Remove contact lenses, if present and easy to do. Continue rinsing.

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Protection of first aiders First aid personnel should wear appropriate protective equipment during any rescue. If it is suspected that volatile contaminants are still present around the affected person, first aid personnel should wear an appropriate respirator or self-contained breathing apparatus. Wash contaminated clothing thoroughly with water before removing it from the affected person, or wear gloves. It may be dangerous for first aid personnel to carry out mouth-to-mouth resuscitation.

4.2. Most important symptoms and effects, both acute and delayed

General information See Section 11 for additional information on health hazards. The severity of the symptoms described will vary dependent on the concentration and the length of exposure.

Inhalation A single exposure may cause the following adverse effects: Dizziness. Headache. May cause respiratory system irritation. Prolonged or repeated exposure may cause the following adverse effects: Nausea, vomiting. Intoxication. Unconsciousness.

Ingestion May cause stomach pain or vomiting. May cause nausea, headache, dizziness and intoxication. Prolonged or repeated exposure may cause the following adverse effects: May cause severe internal injury. Unconsciousness, possibly death.

Skin contact Irritation. Redness. May be absorbed through the skin. Toxic in contact with skin.

Eye contact Causes serious eye damage. Pain. Redness.

4.3. Indication of any immediate medical attention and special treatment needed

Notes for the doctor Treat symptomatically. Keep affected person under observation.

SECTION 5: Firefighting measures

5.1. Extinguishing media

Suitable extinguishing media The product is flammable. Extinguish with alcohol-resistant foam, carbon dioxide, dry powder or water fog. Use fire-extinguishing media suitable for the surrounding fire.

Unsuitable extinguishing media Do not use water jet as an extinguisher, as this will spread the fire.

5.2. Special hazards arising from the substance or mixture

Specific hazards Containers can burst violently or explode when heated, due to excessive pressure build-up. Flammable liquid and vapour. Vapours may be ignited by a spark, a hot surface or an ember. Vapours may form explosive mixtures with air. Fire-water run-off in sewers may create fire or explosion hazard. This product is toxic.

Hazardous combustion products Thermal decomposition or combustion products may include the following substances: Toxic gases or vapours. Oxides of carbon. Aldehydes Formic acid

5.3. Advice for firefighters

Protective actions during firefighting Avoid breathing fire gases or vapours. Evacuate area. Keep upwind to avoid inhalation of gases, vapours, fumes and smoke. Ventilate closed spaces before entering them. Cool containers exposed to heat with water spray and remove them from the fire area if it can be done without risk. Cool containers exposed to flames with water until well after the fire is out. If a leak or spill has not ignited, use water spray to disperse vapours and protect men stopping the leak. Control run-off water by containing and keeping it out of sewers and watercourses. If risk of water pollution occurs, notify appropriate authorities.

Special protective equipment for firefighters Wear chemical protective suit. Wear positive-pressure self-contained breathing apparatus (SCBA) and appropriate protective clothing. Firefighter's clothing conforming to European standard EN469 (including helmets, protective boots and gloves) will provide a basic level of protection for chemical incidents.

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SECTION 6: Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

Personal precautions

No action shall be taken without appropriate training or involving any personal risk. Keep unnecessary and unprotected personnel away from the spillage. Wear protective clothing as described in Section 8 of this safety data sheet. Follow precautions for safe handling described in this safety data sheet. Wash thoroughly after dealing with a spillage. Ensure procedures and training for emergency decontamination and disposal are in place. Do not touch or walk into spilled material. Evacuate area. Provide adequate ventilation. No smoking, sparks, flames or other sources of ignition near spillage. Promptly remove any clothing that becomes contaminated. Avoid inhalation of vapours and spray/mists. Use suitable respiratory protection if ventilation is inadequate. Avoid contact with skin and eyes. Avoid contact with contaminated tools and objects.

6.2. Environmental precautions

Environmental precautions

Avoid discharge into drains or watercourses or onto the ground. Avoid discharge to the aquatic environment. Large Spillages: Inform the relevant authorities if environmental pollution occurs (sewers, waterways, soil or air).

6.3. Methods and material for containment and cleaning up

Methods for cleaning up

Wear protective clothing as described in Section 8 of this safety data sheet. Clear up spills immediately and dispose of waste safely. Eliminate all ignition sources if safe to do so. No smoking, sparks, flames or other sources of ignition near spillage. Use only non-sparking tools. Use explosion-proof electrical equipment. Do not allow material to enter confined spaces, due to the risk of explosion. Provide adequate ventilation. Approach the spillage from upwind. Small Spillages: If the product is soluble in water, dilute the spillage with water and mop it up. Alternatively, or if it is not water-soluble, absorb the spillage with an inert, dry material and place it in a suitable waste disposal container. Large Spillages: If leakage cannot be stopped, evacuate area. Flush spilled material into an effluent treatment plant, or proceed as follows. Contain and absorb spillage with sand, earth or other non-combustible material. Place waste in labelled, sealed containers. Clean contaminated objects and areas thoroughly, observing environmental regulations. The contaminated absorbent may pose the same hazard as the spilled material. Flush contaminated area with plenty of water. Wash thoroughly after dealing with a spillage. Dispose of waste to licensed waste disposal site in accordance with the requirements of the local Waste Disposal Authority.

6.4. Reference to other sections

Reference to other sections

For personal protection, see Section 8. See Section 11 for additional information on health hazards. See Section 12 for additional information on ecological hazards. For waste disposal, see Section 13.

SECTION 7: Handling and storage

7.1. Precautions for safe handling

Usage precautions

Read and follow manufacturer's recommendations. Wear protective clothing as described in Section 8 of this safety data sheet. Keep away from food, drink and animal feeding stuffs. Handle all packages and containers carefully to minimise spills. Keep container tightly sealed when not in use. Avoid the formation of mists. The product is flammable. Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. In use may form flammable/explosive vapour-air mixture. Vapours may accumulate on the floor and in low-lying areas. Use explosion-proof electrical, ventilating and lighting equipment. Use only non-sparking tools. Take precautionary measures against static discharges. This product is toxic. Immediate first aid is imperative. Do not handle until all safety precautions have been read and understood. Do not handle broken packages without protective equipment. Do not reuse empty containers.

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Advice on general occupational hygiene

Wash promptly if skin becomes contaminated. Take off contaminated clothing. Wash contaminated clothing before reuse. Do not eat, drink or smoke when using this product. Wash at the end of each work shift and before eating, smoking and using the toilet. Change work clothing daily before leaving workplace.

7.2. Conditions for safe storage, including any incompatibilities

Storage precautions

Store away from incompatible materials (see Section 10). Store in accordance with local regulations. Eliminate all sources of ignition. Take precautionary measures against static discharges. Earth container and transfer equipment to eliminate sparks from static electricity. Keep away from oxidising materials, heat and flames. Keep only in the original container. Keep container tightly closed, in a cool, well ventilated place. Keep containers upright. Protect containers from damage. Bund storage facilities to prevent soil and water pollution in the event of spillage. The storage area floor should be leak-tight, jointless and not absorbent.

Storage class

Flammable liquid storage.

7.3. Specific end use(s)

Specific end use(s)

The identified uses for this product are detailed in Section 1.2.

SECTION 8: Exposure controls/Personal protection

8.1. Control parameters

Occupational exposure limits

BUTANOL-norm

Long-term exposure limit (8-hour TWA): WEL

Short-term exposure limit (15-minute): WEL 50 ppm(Sk) 154 mg/m³(Sk)

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Long-term exposure limit (8-hour TWA): WEL 200 ppm(Sk) 266 mg/m³(Sk)

Short-term exposure limit (15-minute): WEL 250 ppm(Sk) 333 mg/m³(Sk)

WEL = Workplace Exposure Limit.

BUTANOL-norm (CAS: 71-36-3)

DNEL

Workers - Inhalation; Long term local effects: 310 mg/m³

General population - Inhalation; Long term systemic effects: 55.357 mg/m³

General population - Inhalation; Long term local effects: 155 mg/m³

General population - Dermal; Long term systemic effects: 3.125 mg/kg/day

General population - Oral; Long term systemic effects: 1.562 mg/kg/day

PNEC

- Fresh water; 0.082 mg/l

- marine water; 0.008 mg/l

- STP; 2476 mg/l

- Sediment (Freshwater); 0.324 mg/kg

- Sediment (Marinewater); 0.032 mg/kg

- Soil; 0.017 mg/kg

METHANOL (CAS: 67-56-1)

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DNEL

Workers - Inhalation; Long term systemic effects: 130 mg/m³
 Workers - Inhalation; Short term systemic effects: 130 mg/m³
 Workers - Inhalation; Long term local effects: 130 mg/m³
 Workers - Inhalation; Short term local effects: 130 mg/m³
 Workers - Dermal; Long term systemic effects: 20 mg/kg/day
 Workers - Dermal; Short term systemic effects: 20 mg/kg/day
 General population - Inhalation; Long term systemic effects: 26 mg/m³
 General population - Inhalation; Short term systemic effects: 26 mg/m³
 General population - Inhalation; Long term local effects: 26 mg/m³
 General population - Inhalation; Short term local effects: 26 mg/m³
 General population - Dermal; Long term systemic effects: 4 mg/kg/day
 General population - Dermal; Short term systemic effects: 4 mg/kg/day
 General population - Oral; Long term systemic effects: 4 mg/kg/day
 General population - Oral; Short term systemic effects: 4 mg/kg/day

PNEC

- Fresh water; 20.8 mg/l
 - marine water; 2.08 mg/l
 - STP; 100 mg/l
 - Sediment (Freshwater); 77 mg/kg
 - Sediment (Marinewater); 7.7 mg/kg
 - Soil; 100 mg/kg

8.2. Exposure controls

Protective equipment



Appropriate engineering controls

Provide adequate ventilation. Personal, workplace environment or biological monitoring may be required to determine the effectiveness of the ventilation or other control measures and/or the necessity to use respiratory protective equipment. Use process enclosures, local exhaust ventilation or other engineering controls as the primary means to minimise worker exposure. Personal protective equipment should only be used if worker exposure cannot be controlled adequately by the engineering control measures. Ensure control measures are regularly inspected and maintained. Ensure operatives are trained to minimise exposure. The engineering controls also need to keep gas, vapour or dust concentrations below any lower explosive limits. Use explosion-proof ventilating equipment.

Eye/face protection

Eyewear complying with an approved standard should be worn if a risk assessment indicates eye contact is possible. Personal protective equipment for eye and face protection should comply with European Standard EN166. Wear tight-fitting, chemical splash goggles or face shield. If inhalation hazards exist, a full-face respirator may be required instead.

Hand protection

Chemical-resistant, impervious gloves complying with an approved standard should be worn if a risk assessment indicates skin contact is possible. To protect hands from chemicals, gloves should comply with European Standard EN374. Considering the data specified by the glove manufacturer, check during use that the gloves are retaining their protective properties and change them as soon as any deterioration is detected. Frequent changes are recommended. It is recommended that gloves are made of the following material: Butyl rubber. Thickness: ≥ 0.3 mm The selected gloves should have a breakthrough time of at least 8 hours. The breakthrough time for any glove material may be different for different glove manufacturers. The most suitable glove should be chosen in consultation with the glove supplier/manufacturer, who can provide information about the breakthrough time of the glove material.

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Other skin and body protection	Appropriate footwear and additional protective clothing complying with an approved standard should be worn if a risk assessment indicates skin contamination is possible.
Hygiene measures	Provide eyewash station and safety shower. Contaminated work clothing should not be allowed out of the workplace. Wash contaminated clothing before reuse. Clean equipment and the work area every day. Good personal hygiene procedures should be implemented. Wash at the end of each work shift and before eating, smoking and using the toilet. When using do not eat, drink or smoke. Preventive industrial medical examinations should be carried out. Warn cleaning personnel of any hazardous properties of the product.
Respiratory protection	Respiratory protection complying with an approved standard should be worn if a risk assessment indicates inhalation of contaminants is possible. Ensure all respiratory protective equipment is suitable for its intended use and is 'CE'-marked. Check that the respirator fits tightly and the filter is changed regularly. Gas and combination filter cartridges should comply with European Standard EN14387. Full face mask respirators with replaceable filter cartridges should comply with European Standard EN136. Half mask and quarter mask respirators with replaceable filter cartridges should comply with European Standard EN140. Organic vapour filter.
Environmental exposure controls	Keep container tightly sealed when not in use. Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels. Store in a demarcated bunded area to prevent release to drains and/or watercourses.

SECTION 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

Appearance	Liquid.
Colour	Colourless.
Odour	Characteristic.
Odour threshold	Not determined.
pH	Not determined.
Melting point	Not determined.
Initial boiling point and range	Not determined.
Flash point	Not determined.
Upper/lower flammability or explosive limits	Not determined.
Vapour pressure	Not determined.
Relative density	~ 0.8
Solubility(ies)	Soluble in water.
Partition coefficient	Not determined.
Auto-ignition temperature	Not determined.
Decomposition Temperature	Not determined.
Viscosity	Not determined.
Explosive properties	Not considered to be explosive.
Oxidising properties	Does not meet the criteria for classification as oxidising.

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9.2. Other information

Other information None.

SECTION 10: Stability and reactivity

10.1. Reactivity

Reactivity The following materials may react with the product: Acids. Oxidising agents.

10.2. Chemical stability

Stability Stable at normal ambient temperatures and when used as recommended. Stable under the prescribed storage conditions.

10.3. Possibility of hazardous reactions

Possibility of hazardous reactions May generate heat. In contact with some metals can generate hydrogen gas, which can form explosive mixtures with air.

10.4. Conditions to avoid

Conditions to avoid Avoid heat, flames and other sources of ignition. Containers can burst violently or explode when heated, due to excessive pressure build-up. Static electricity and formation of sparks must be prevented. Do not pressurise, cut, weld, drill, grind or otherwise expose containers to heat or sources of ignition.

10.5. Incompatible materials

Materials to avoid Oxidising materials. Acids. Chemically-active metals. Aluminium. Zinc

10.6. Hazardous decomposition products

Hazardous decomposition products Does not decompose when used and stored as recommended.

SECTION 11: Toxicological information

11.1. Information on toxicological effects

Acute toxicity - oral

Notes (oral LD₅₀) Acute Tox. 3 - H301 Toxic if swallowed.

ATE oral (mg/kg) 201.61

Acute toxicity - dermal

Notes (dermal LD₅₀) Acute Tox. 3 - H311 Toxic in contact with skin.

ATE dermal (mg/kg) 810.81

Acute toxicity - inhalation

Notes (inhalation LC₅₀) Based on available data the classification criteria are not met.

Skin corrosion/irritation

Animal data Irritating.

Serious eye damage/irritation

Serious eye damage/irritation Eye Dam. 1 - H318 Causes serious eye damage.

Respiratory sensitisation

Respiratory sensitisation Based on available data the classification criteria are not met.

Skin sensitisation

Skin sensitisation Based on available data the classification criteria are not met.

Germ cell mutagenicity

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Genotoxicity - in vitro	Based on available data the classification criteria are not met.
<u>Carcinogenicity</u>	
Carcinogenicity	Based on available data the classification criteria are not met.
IARC carcinogenicity	None of the ingredients are listed or exempt.
<u>Reproductive toxicity</u>	
Reproductive toxicity - fertility	Based on available data the classification criteria are not met.
Reproductive toxicity - development	Based on available data the classification criteria are not met.
<u>Specific target organ toxicity - single exposure</u>	
STOT - single exposure	STOT SE 3 - H335, H336 May cause respiratory irritation. May cause drowsiness or dizziness. STOT SE 1 - H370 Causes damage to organs .
Target organs	Respiratory system, lungs Central nervous system
<u>Specific target organ toxicity - repeated exposure</u>	
STOT - repeated exposure	Not classified as a specific target organ toxicant after repeated exposure.
<u>Aspiration hazard</u>	
Aspiration hazard	Based on available data the classification criteria are not met.
<u>General information</u>	
	The severity of the symptoms described will vary dependent on the concentration and the length of exposure.
Inhalation	A single exposure may cause the following adverse effects: Irritation. Headache. Prolonged or repeated exposure may cause the following adverse effects: May cause drowsiness or dizziness. Nausea, vomiting. Intoxication. Muscle weakness. Unconsciousness.
Ingestion	Stomach pain. Nausea, vomiting. Prolonged or repeated exposure may cause the following adverse effects: May cause severe internal injury. Unconsciousness, possibly death.
Skin contact	Irritation. May be absorbed through the skin. Toxic in contact with skin.
Eye contact	Causes serious eye damage. Symptoms following overexposure may include the following: Pain. Profuse watering of the eyes. Redness.
Route of exposure	Ingestion Inhalation Skin and/or eye contact
Target organs	Central nervous system Respiratory system, lungs
<u>Toxicological information on ingredients.</u>	

BUTANOL-norm

Acute toxicity - oral

Acute toxicity oral (LD₅₀ mg/kg) 2,920.0

Species Rat

ATE oral (mg/kg) 500.0

Acute toxicity - dermal

Acute toxicity dermal (LD₅₀ mg/kg) 3,430.0

Species Rabbit

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Acute toxicity - inhalation

Notes (inhalation LC₅₀) LC0 17.7 mg/l, 4 hours, Vapour Rat

Skin corrosion/irritation

Animal data Erythema/eschar score: 1.7 to 4 Oedema score: 0.7 to 2 Irritating.

Serious eye damage/irritation

Serious eye damage/irritation Irritating Tests on rabbits, OECD Guideline 405, Acute eye Irritation / Corrosion.

Respiratory sensitisation

Respiratory sensitisation Based on available data the classification criteria are not met.

Skin sensitisation

Skin sensitisation Based on available data the classification criteria are not met.

Germ cell mutagenicity

Genotoxicity - in vitro Gene mutation:: Negative., With and without metabolic activation.

Genotoxicity - in vivo Gene mutation: Negative.

Carcinogenicity

Carcinogenicity Based on available data the classification criteria are not met.

Reproductive toxicity

Reproductive toxicity - fertility Two-generation study - LOAEC 750 ppm, Inhalation, Rat P

Reproductive toxicity - development Maternal toxicity: - NOAEL: 1454 mg/kg/day, Oral, Rat

Specific target organ toxicity - repeated exposure

STOT - repeated exposure LOAEL 500 mg/kg/day, Oral, Rat 6 weeks

Inhalation Irritating to respiratory system.

Ingestion Harmful if swallowed. May cause nausea, headache, dizziness and intoxication. May cause stomach pain or vomiting.

Skin contact Irritating to skin. Redness. Prolonged contact may cause dryness of the skin.

Eye contact Irritating to eyes.

Route of exposure Inhalation Skin and/or eye contact Ingestion

Target organs Eyes Skin Central nervous system Respiratory system, lungs

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Acute toxicity - oral

Notes (oral LD₅₀) LD0 2528 mg/kg, Oral, Rat

ATE oral (mg/kg) 100.0

Acute toxicity - dermal

Notes (dermal LD₅₀) No information available.

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ATE dermal (mg/kg)	300.0
<u>Acute toxicity - inhalation</u>	
Acute toxicity inhalation (LC ₅₀ vapours mg/l)	43.68
Species	Rat
Notes (inhalation LC ₅₀)	6 hours REACH dossier information.
ATE inhalation (vapours mg/l)	3.0
<u>Skin corrosion/irritation</u>	
Human skin model test	No information available.
<u>Serious eye damage/irritation</u>	
Serious eye damage/irritation	Based on available data the classification criteria are not met.
<u>Respiratory sensitisation</u>	
Respiratory sensitisation	No specific test data are available.
<u>Skin sensitisation</u>	
Skin sensitisation	Guinea pig maximization test (GPMT) - Guinea pig: Not sensitising.
<u>Germ cell mutagenicity</u>	
Genotoxicity - in vitro	Bacterial reverse mutation test: Negative.
Genotoxicity - in vivo	Micronucleus assay: Negative.
<u>Carcinogenicity</u>	
Carcinogenicity	NOAEC >1.3 mg/l, Inhalation, Mouse
<u>Reproductive toxicity</u>	
Reproductive toxicity - fertility	Fertility - LOAEC 1000 mg/kg, Oral, Mouse P
Reproductive toxicity - development	Developmental toxicity: - NOAEL: 5000 mg/l, Oral, Mouse
<u>Specific target organ toxicity - single exposure</u>	
STOT - single exposure	Conclusive data but not sufficient for classification.
Target organs	Central nervous system
<u>Specific target organ toxicity - repeated exposure</u>	
STOT - repeated exposure	LOAEC 1.3 mg/l, Inhalation, Rat
Target organs	Central nervous system Eyes
<u>Aspiration hazard</u>	
Aspiration hazard	No information available.

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Inhalation	Toxic by inhalation. A single exposure may cause the following adverse effects: Vapours may cause headache, fatigue, dizziness and nausea. Prolonged or repeated exposure may cause the following adverse effects: Central nervous system depression.
Ingestion	Toxic if swallowed. A single exposure may cause the following adverse effects: May cause nausea, headache, dizziness and intoxication. Prolonged or repeated exposure may cause the following adverse effects: Central nervous system depression. Unconsciousness, possibly death.
Skin contact	Toxic in contact with skin. May be absorbed through the skin. Prolonged skin contact may cause redness and irritation.
Eye contact	Irritating to eyes.
Route of exposure	Inhalation Skin absorption Ingestion.
Target organs	Central nervous system Eyes

SECTION 12: Ecological information

Ecotoxicity	Not regarded as dangerous for the environment. However, large or frequent spills may have hazardous effects on the environment.
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12.1. Toxicity

Toxicity	Based on available data the classification criteria are not met.
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Ecological information on ingredients.

BUTANOL-norm

Acute aquatic toxicity

Acute toxicity - fish	LC ₅₀ , 96 hours: 1376 mg/l, Pimephales promelas (Fat-head Minnow)
Acute toxicity - aquatic invertebrates	EC ₅₀ , 48 hours: 1328 mg/l, Daphnia magna
Acute toxicity - aquatic plants	EC ₅₀ , 96 hours: 225 mg/l, Pseudokirchneriella subcapitata
Acute toxicity - microorganisms	EC ₅₀ , 17 hours: 4390 mg/l, Pseudomonas putida

Chronic aquatic toxicity

Chronic toxicity - fish early life stage	Not available.
Short term toxicity - embryo and sac fry stages	Not available.
Chronic toxicity - aquatic invertebrates	EC ₅₀ , 21 days: 18 mg/l, Daphnia magna

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Acute aquatic toxicity

Acute toxicity - fish	LC ₅₀ , 96 hours: 15400 mg/l, Lepomis macrochirus (Bluegill)
Acute toxicity - aquatic invertebrates	EC ₅₀ , 96 hours: 18260 mg/l, Daphnia magna

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Acute toxicity - aquatic plants	EC ₅₀ , 96 hours: 22000 mg/l, <i>Pseudokirchneriella subcapitata</i>
Acute toxicity - microorganisms	EC ₅₀ , 15 hours: 20000 mg/l, Activated sludge
Acute toxicity - terrestrial	NOEC, 35 days: 10000 mg/kg,
<u>Chronic aquatic toxicity</u>	
Chronic toxicity - fish early life stage	EC ₅₀ , 200 hours: 14536 mg/l, <i>Oryzias latipes</i> (Red killifish)

12.2. Persistence and degradability

Persistence and degradability Expected to be readily biodegradable.

Ecological information on ingredients.

BUTANOL-norm

Phototransformation	Water - DT ₅₀ : 46-35.5 hours
Stability (hydrolysis)	Scientifically unjustified.
Biodegradation	The substance is readily biodegradable. Water - Degradation (%) 92%: 20 days

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Phototransformation	Air - Degradation (%) 50: 17.2 days
Stability (hydrolysis)	Scientifically unjustified.
Biodegradation	Water - Degradation 82.7: 5 days The substance is readily biodegradable.

Biological oxygen demand 1.236 g O₂/g substance

12.3. Bioaccumulative potential

Bioaccumulative potential Bioaccumulation is unlikely.

Partition coefficient Not determined.

Ecological information on ingredients.

BUTANOL-norm

Bioaccumulative potential	Bioaccumulation is unlikely.
Partition coefficient	Pow: 10 @ 25°C and pH7

METHANOL

Bioaccumulative potential	Scientifically unjustified. BCF: 1,
Partition coefficient	log Pow: -0.77 REACH dossier information.

12.4. Mobility in soil

Mobility The product is water-soluble and may spread in water systems. The product contains volatile substances which may spread in the atmosphere.

Ecological information on ingredients.

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BUTANOL-norm

Mobility	The product is miscible with water and may spread in water systems.
Adsorption/desorption coefficient	Soil - Log Koc: 0.54 @ °C
Henry's law constant	0.054 Pa m ³ /mol @ °C
Surface tension	69.9 mN/m @ 20°C Concentration = 1g/l

METHANOL

Mobility	The product is miscible with water and may spread in water systems.
Adsorption/desorption coefficient	Expected to have a low potential for adsorption.
Henry's law constant	0.461 Pa m ³ /mol @ 25°C
Surface tension	No specific test data are available.

12.5. Results of PBT and vPvB assessment

Results of PBT and vPvB assessment This product does not contain any substances classified as PBT or vPvB.

Ecological information on ingredients.

BUTANOL-norm

Results of PBT and vPvB assessment This substance is not classified as PBT or vPvB according to current EU criteria.

METHANOL

Results of PBT and vPvB assessment This substance is not classified as PBT or vPvB according to current EU criteria.

12.6. Other adverse effects

Other adverse effects None known.

Ecological information on ingredients.

BUTANOL-norm

Other adverse effects None known.

METHANOL

Other adverse effects None known.

SECTION 13: Disposal considerations

13.1. Waste treatment methods

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General information

The generation of waste should be minimised or avoided wherever possible. Reuse or recycle products wherever possible. This material and its container must be disposed of in a safe way. Disposal of this product, process solutions, residues and by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any local authority requirements. When handling waste, the safety precautions applying to handling of the product should be considered. Care should be taken when handling emptied containers that have not been thoroughly cleaned or rinsed out. Empty containers or liners may retain some product residues and hence be potentially hazardous.

Disposal methods

Do not empty into drains. Dispose of surplus products and those that cannot be recycled via a licensed waste disposal contractor. Waste, residues, empty containers, discarded work clothes and contaminated cleaning materials should be collected in designated containers, labelled with their contents. Incineration or landfill should only be considered when recycling is not feasible. Vapour from residual product may create a highly flammable or explosive atmosphere inside the container. Containers should be thoroughly emptied before disposal because of the risk of an explosion. Do not cut or weld used containers unless they have been thoroughly cleaned internally.

SECTION 14: Transport information

General

For limited quantity packaging/limited load information, consult the relevant modal documentation using the data shown in this section.

14.1. UN number

UN No. (ADR/RID)	1986
UN No. (IMDG)	1986
UN No. (ICAO)	1986

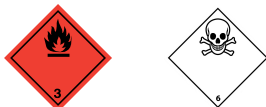
14.2. UN proper shipping name

Proper shipping name (ADR/RID)	ALCOHOLS, FLAMMABLE, TOXIC, N.O.S. (BUTAN-1-OL, METHANOL)
Proper shipping name (IMDG)	ALCOHOLS, FLAMMABLE, TOXIC, N.O.S. (BUTAN-1-OL, METHANOL)
Proper shipping name (ICAO)	ALCOHOLS, FLAMMABLE, TOXIC, N.O.S. (BUTAN-1-OL, METHANOL)
Proper shipping name (ADN)	ALCOHOLS, FLAMMABLE, TOXIC, N.O.S. (BUTAN-1-OL, METHANOL)

14.3. Transport hazard class(es)

ADR/RID class	3
ADR/RID subsidiary risk	6.1
ADR/RID label	3 & 6.1
IMDG class	3
IMDG subsidiary risk	6.1
ICAO class/division	3
ICAO subsidiary risk	6.1

Transport labels



14.4. Packing group

MIXED ALCOHOL SOLVENT

ADR/RID packing group II

IMDG packing group II

ICAO packing group II

14.5. Environmental hazards

Environmentally hazardous substance/marine pollutant

No.

14.6. Special precautions for user

Always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.

EmS F-E, S-D

Emergency Action Code 3WE

Hazard Identification Number 336
(ADR/RID)

14.7. Transport in bulk according to Annex II of MARPOL and the IBC Code

Transport in bulk according to Not applicable.

Annex II of MARPOL 73/78
and the IBC Code

SECTION 15: Regulatory information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

National regulations Health and Safety at Work etc. Act 1974 (as amended).
The Carriage of Dangerous Goods and Use of Transportable Pressure Equipment
Regulations 2009 (SI 2009 No. 1348) (as amended) ["CDG 2009"].
EH40/2005 Workplace exposure limits.
UK REACH and UK CLP Regulations.

15.2. Chemical safety assessment

No chemical safety assessment has been carried out.

Inventories

EU - EINECS/ELINCS

None of the ingredients are listed or exempt.

SECTION 16: Other information

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Abbreviations and acronyms used in the safety data sheet	ADR: European Agreement concerning the International Carriage of Dangerous Goods by Road. ADN: European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways. RID: European Agreement concerning the International Carriage of Dangerous Goods by Rail. IATA: International Air Transport Association. ICAO: Technical Instructions for the Safe Transport of Dangerous Goods by Air. IMDG: International Maritime Dangerous Goods. CAS: Chemical Abstracts Service. ATE: Acute Toxicity Estimate. LC₅₀: Lethal Concentration to 50 % of a test population. LD₅₀: Lethal Dose to 50% of a test population (Median Lethal Dose). EC₅₀: 50% of maximal Effective Concentration. PBT: Persistent, Bioaccumulative and Toxic substance. vPvB: Very Persistent and Very Bioaccumulative.
Classification abbreviations and acronyms	Flam. Liq. = Flammable liquid Acute Tox. = Acute toxicity Eye Dam. = Serious eye damage Skin Irrit. = Skin irritation STOT SE = Specific target organ toxicity-single exposure
General information	This datasheet is not intended to be a replacement for a full risk assessment, these should always be carried out by competent persons.
Key literature references and sources for data	Source: European Chemicals Agency, http://echa.europa.eu/
Classification procedures according to Regulation (EC) 1272/2008	Acute Tox. 3 - H311: Acute Tox. 3 - H301: Eye Dam. 1 - H318: STOT SE 1 - H370: STOT SE 3 - H335, H336: Skin Irrit. 2 - H315: : Calculation method. Flam. Liq. 2 - H225: : Expert judgement.
Training advice	Only trained personnel should use this material.
Revision comments	General review
Revision date	15/04/2021
Revision	1
Supersedes date	01/10/2005
SDS number	10883
Hazard statements in full	H225 Highly flammable liquid and vapour. H226 Flammable liquid and vapour. H301 Toxic if swallowed. H302 Harmful if swallowed. H311 Toxic in contact with skin. H315 Causes skin irritation. H318 Causes serious eye damage. H331 Toxic if inhaled. H335 May cause respiratory irritation. H336 May cause drowsiness or dizziness. H370 Causes damage to organs (Central nervous system, Eyes). H370 Causes damage to organs .

MIXED ALCOHOL SOLVENT

This information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process. Such information is, to the best of the company's knowledge and belief, accurate and reliable as of the date indicated. However, no warranty, guarantee or representation is made to its accuracy, reliability or completeness. It is the user's responsibility to satisfy himself as to the suitability of such information for his own particular use.